

PAPER_1409_QUARK_GENERATIONS_THREE

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PAPER_1409 — UQFF Derivation of Three Quark/Lepton Generations (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: D — Particle Physics (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: `calculate_particle_physics({'observable':'n_generations'})` →
`D_phys - 1 = 3 EXACT`

Observation

Why exactly three generations of quarks and leptons? Standard Model takes this as observed input; LEP confirmed $N = 2.984 \pm 0.008$ from Z-width.

UQFF Closed Identity

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`N_generations = D_phys - 1 = 3 EXACT`

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Physical Interpretation

Spacetime dimensionality $D_{\text{phys}} = 4$ determines the fermion-generation count via $SO(D_{\text{phys}}-1)$ flavor symmetry. The fourth generation would require $D_{\text{phys}} = 5$, contradicting observed Minkowski structure.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF EXACT integer-primitive identity for Three Quark/Lepton Generations.